



7213 - Probing the disk evolution in an early stage massive protocluster

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC6334I(N)				
	1	NIRCam Imaing	NIRCam Imaging	(1) NGC6334_I_N_NIRCam
	2	MIRI Imaging	MIRI Imaging	(2) NGC6334_I_N-MIRI
	3	MIRI Imaging	MIRI Imaging	(3) NGC6334_I_N_bkg

ABSTRACT

Circumstellar disks are critical to the formation of stars and planets, yet most surveys of these disks focus on nearby, low-mass star-forming regions. This bias may limit our understanding of the more typical environments for planet formation, as most stars form in dense clusters. Extending these studies to distant, massive protoclusters has mainly been restricted by the lack of prior YSO identification and classification from infrared surveys. We propose to utilize the unprecedented sensitivity and resolution of JWST to conduct the first deep census of YSOs in the early-stage massive protocluster NGC6334I(N) and NGC6334I. Our goal is to identify YSOs down to $0.1L_{\text{sun}}$ for $AK < 10$ and classify the evolution classes of cluster members with near- to mid-infrared photometry. The survey will yield a sample of over 10^3 YSOs, achieving a depth comparable to infrared surveys in nearby regions. Combined with joint ALMA observation (15au resolution, $0.87\sim\text{mm}$) in a pilot sample of over 200 disks, we will measure the distribution of dust masses and radii for disks at different evolutionary phases and search for disk substructures. This will provide new insights regarding how disk evolution unfolds in early-stage massive clusters, with potential implications for our understanding of star and planet formation on a broader scale.

OBSERVING DESCRIPTION

We plan to use a 2×1 NIRCcam mosaic and a 6×2 MIRI mosaic to observe the most intense star-forming clump N6334I and N6334I(N) in a $6'\times 4'$ FOV. For NIRCcam we will observe F115W, F162M, F182M, F200W, F356W, F405N, F444W, F470N via 4 SW+LW pairings. This setup includes wide/medium filters that match existing stellar populations work employing standard Johnson J/H/Ks and Spitzer IRAC 3.6/4.5 μm for SED study; and also narrow filters F405N to observe Br as a disk accretion tracer, F470N as an outflow tracer. Five MIRI filters, F770W, F1000W, F1500W, F2100W and F2500W, are chosen to sample the SED of the sources in the longer wavelengths.

Proposal 7213 - Targets - Probing the disk evolution in an early stage massive protocluster

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC6334_I_N_NIRCam	RA: 17 20 54.3648 (260.2265200d) Dec: -35 46 0.72 (-35.76687d) Equinox: J2000	Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Stellar Cluster</i> <i>Description=[Protoclusters, Young star clusters]</i> <i>Extended=YES</i>				
	(2)	NGC6334_I_N-MIRI	RA: 17 20 54.3648 (260.2265200d) Dec: -35 46 0.72 (-35.76687d) Equinox: J2000	Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Stellar Cluster</i> <i>Description=[Protoclusters, Young star clusters]</i> <i>Extended=YES</i>				
	(3)	NGC6334_I_N_bkg	RA: 17 21 17.4257 (260.3226071d) Dec: -35 40 51.58 (-35.68099d) Equinox: J2000	Epoch of Position: 2000	
	<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Stellar Cluster</i> <i>Description=[Young star clusters]</i> <i>Extended=YES</i>				

Proposal 7213 - Observation 1 - Probing the disk evolution in an early stage massive protocluster

Observation	Proposal 7213, Observation 1: NIRCam Imaing									Tue Dec 09 16:00:22 GMT 2025
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(NIRCam Imaing (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(Visit 1:1) Warning (Form): Data Excess over lower threshold									
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:2) Warning (Form): Data Excess over lower threshold									
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.									
(Visit 1:2) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	NGC6334_I_N_NIRCam	RA: 17 20 54.3648 (260.2265200d) Dec: -35 46 0.72 (-35.76687d) Equinox: J2000			Epoch of Position: 2000				
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.									
	Category=Stellar Cluster Description=[Protoclusters, Young star clusters] Extended=YES									
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)	Tile Order
	2	1	10.0		10.0		0.0		0.0	DEFAULT
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		5TIGHT		STANDARD				2
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F162M+F150W2	F405N+F444W	BRIGHT2	6	1	10	10	1288.412	
	2	F182M	F470N+F444W	BRIGHT2	6	1	10	10	1288.412	
	3	F115W	F356W	BRIGHT2	5	1	10	10	1073.677	
	4	F200W	F444W	BRIGHT2	5	1	10	10	1073.677	

Proposal 7213 - Observation 1 - Probing the disk evolution in an early stage massive protocluster

Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 90 to 130 Degrees (V3 90.07457694 to 130.07457694) Aperture PA Range 260 to 300 Degrees (V3 260.07457694 to 300.07457694) Visits Same PA Same Aperture PA 1, 2 (V3 PAs differ)
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Proposal 7213 - Observation 2 - Probing the disk evolution in an early stage massive protocluster

Tue Dec 09 16:00:22 GMT 2025

Observation	Proposal 7213, Observation 2: MIRI Imaging										Tue Dec 09 16:00:22 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observations:[MIRI Imaging (Obs 3)]										
Diagnostics	(MIRI Imaging (Obs 2)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 2:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(MIRI Imaging (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.										
	(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 2:2) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 2:3) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 2:4) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 2:5) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	(Visit 2:6) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	(Visit 2:7) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(2)	NGC6334_I_N-MIRI	RA: 17 20 54.3648 (260.2265200d)			Epoch of Position: 2000					
			Dec: -35 46 0.72 (-35.76687d)								
			Equinox: J2000								
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
Category=Stellar Cluster											
Description=[Protoclusters, Young star clusters]											
Extended=YES											
Template	Subarray										
	FULL										
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)	Column shift (deg)	Tile Order		
	2	5	10.0		10.0		0.0	0.0	DEFAULT		
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	309	5						LARGE	

Proposal 7213 - Observation 2 - Probing the disk evolution in an early stage massive protocluster

Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	20	1	1	Dither 1	5	5	277.504	
	2	F1000W	FASTR1	20	1	1	Dither 1	5	5	277.504	
	3	F1500W	FASTR1	20	1	1	Dither 1	5	5	277.504	
	4	F2100W	FASTR1	20	1	1	Dither 1	5	5	277.504	
	5	F2550W	FASTR1	20	1	1	Dither 1	5	5	277.504	
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 90 to 130 Degrees (V3 85.16455103 to 125.16455103) Aperture PA Range 260 to 300 Degrees (V3 255.16455103 to 295.16455103) Visits Same PA										
	Sequence Observations 3, 2 (reordered), Non-interruptible Same Aperture PA 1, 2 (V3 PAs differ)										

Proposal 7213 - Observation 3 - Probing the disk evolution in an early stage massive protocluster

Observation	Proposal 7213, Observation 3: MIRI Imaging										Tue Dec 09 16:00:22 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
	Background Observation For: [MIRI Imaging (Obs 2)]										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 3:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	NGC6334_I_N_bkg	RA: 17 21 17.4257 (260.3226071d) Dec: -35 40 51.58 (-35.68099d) Equinox: J2000			Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Stellar Cluster Description=[Young star clusters] Extended=YES										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	2-Point	309	5						LARGE	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F770W	FASTR1	20	1	1	Dither 1	2	2	111.002	
	2	F1000W	FASTR1	20	1	1	Dither 1	2	2	111.002	
	3	F1500W	FASTR1	20	1	1	Dither 1	2	2	111.002	
	4	F2100W	FASTR1	20	1	1	Dither 1	2	2	111.002	
	5	F2550W	FASTR1	20	1	1	Dither 1	2	2	111.002	
Special Requirements	Aperture PA Range 90 to 130 Degrees (V3 85.16455103 to 125.16455103) Aperture PA Range 260 to 300 Degrees (V3 255.16455103 to 295.16455103)										
	Sequence Observations 3, 2 (reordered), Non-interruptible										